

WASP-35b

R-1233

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-35_240102 · created 2026-07-31T10:05:57+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460311.7652 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.048
Transit depth (Rp/Rs)^2	2.45 +/- 1.49 [%]
Orbital Inclination (inc)	89.62 +/- 1.99
Airmass coefficient 1 (a1)	0.89 +/- 0.22
Airmass coefficient 2 (a2)	0.11 +/- 0.39
Scatter in the residuals of the lightcurve fit is	24.74 %
Best Comparison Star	#2 - [222, 292]
Optimal Aperture	1.25
Optimal Annulus	3.68
Transit Duration (day)	0.1247 +/- 0.0069

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.156 ± 0.048 vs 0.1241 (deviation 0.66σ)
Duration fit vs published	0.1247 d vs 0.1317 d (deviation 1.01σ)
Mid-time O–C	-9.2 min
Depth SNR	3.2
Residual scatter	24.74 %

Light curve

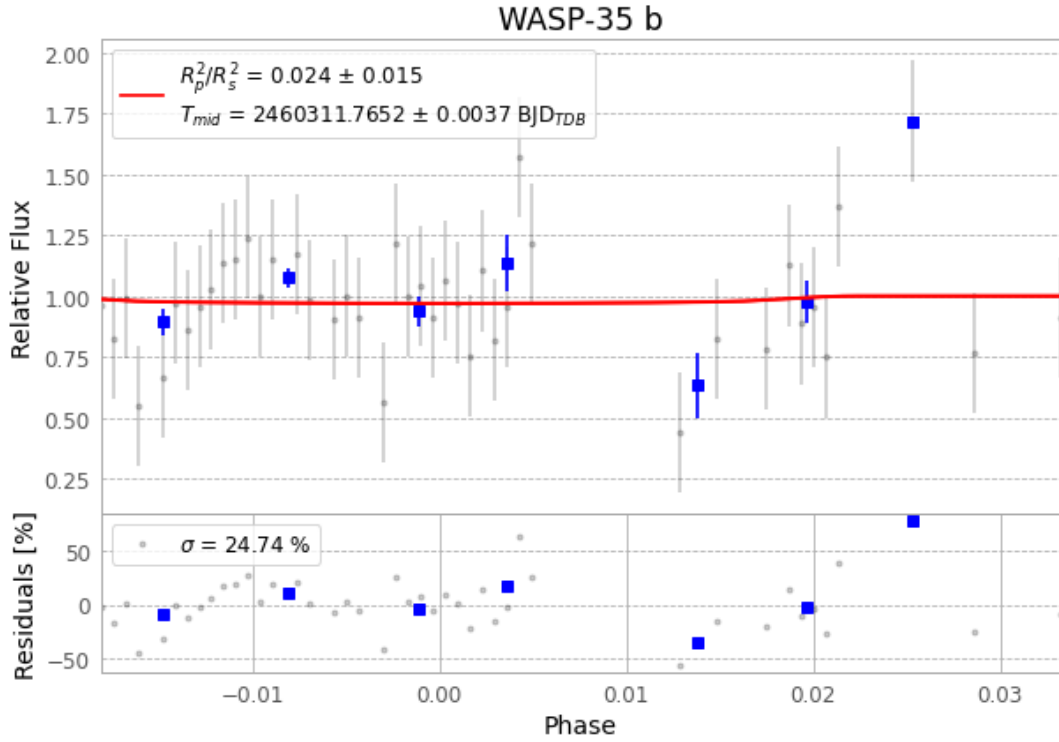


Figure 1 — FinalLightCurve_WASP-35 b_2024-01-01.png (SHA-256 0b481fe3a8d61958...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [406, 231], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 da055881abbf09fb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

83 FITS frames (55 MB), WASP-35240102045116.FITS → WASP-35240102085715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-240102.log (adf81bde26a0...), FinalLightCurve_WASP-35 b_2024-01-01.png (0b481fe3a8d6...), FinalLightCurve_WASP-35 b_2024-01-01.csv (6fab48bb3eef...)

Compute resources

Wall time 190.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 10:05Z.